Telmo Cunha

Website | LinkedIn | GitHub

Research Experience

- **Research Assistant at CeFEMA - Instituto Superior Técnico** — Lisbon, Portugal
  Topic: Nonequilibrium quantum many-body dynamics and Hilbert space fragmentation — *03/2026 - 10/2026*
  - Developed and implemented Python-based numerical simulations of classical and quantum many-body spin systems with constrained dynamics, investigating phase transitions associated with Hilbert space fragmentation.
- **Research Assistant at Aalto University (AScI Program)** — Espoo, Finland
  Topic: Equivariance properties of Graph Neural Networks (GNNs) — *04/2023 - 09/2023*
  - Reviewed the scientific literature on GNNs and geometric deep learning. Researched expressivity properties of equivariant GNNs to physical symmetries in molecular dynamics.

Education

- **MSc in Applied Mathematics and Computation - Instituto Superior Técnico** — Lisbon, Portugal
  Final grade: 17/20 — *2025*
  - Specialization in Pure Mathematics, with elective courses in Mathematical Physics and Machine Learning.
  - Thesis: Semilinear Wave Equations on Decelerated Expanding FLRW Spacetimes (Grade: 19/20).
- **BSc in Engineering Physics - Instituto Superior Técnico** — Lisbon, Portugal
  Final grade: 15/20 — *2019*
  - Specialization in theoretical physics.
  - Exchange semester in theoretical physics at EPFL in Lausanne, Switzerland.

Teaching Experience

- **Invited Assistant Professor at ISCTE – Instituto Universitário de Lisboa** — Lisbon, Portugal
  Taught Calculus I, Calculus II and Linear Algebra to first-year Economics students. — *2024, 2025*
- **Teaching Assistant at ISCTE – Instituto Universitário de Lisboa** — Lisbon, Portugal
  Taught review lectures of Calculus I to international students in Management (in English). — *2024, 2025*
- **Tutor in Mathematics and Physics** — Lisbon, Portugal
  Individual instruction in mathematics and physics to high-school and undergraduate students. — *2015 – 2023*

Technical Skills

- **Programming**: Python (proficient), C/C++ and R (academic experience)
- **Python Libraries**: NumPy, SciPy, pandas, scikit-learn, PyTorch, matplotlib
- **Tools**: Mathematica, LaTeX, Git

Selected Computational Projects

- **Mathematical Foundations of Machine Learning**: Developed theoretical notes on learning theory, machine learning and kernel methods, with Python implementations (GitHub).
- **Manifold Learning**: Implemented manifold learning and nonlinear optimisation methods for dimensionality reduction (GitHub).
- **Feature Selection**: Implemented information-theoretic feature-selection methods for dimensionality reduction and predictive modelling (GitHub).
- **Collaborative Filtering**: Implemented a matrix-factorisation-based movie recommendation system (GitHub).

Additional Training

- **Quantum Landscapes International School** — Paris, France
  International school on nonequilibrium quantum many-body physics. — *2026*

Language Skills

- **Languages**: Portuguese (Native); English (Fluent); Spanish (Basic); French (Basic);